

Automatic non-halting certificates for the Space Needle

WS1, first execution — an impossibility theorem at bounded size, with audited witnesses — July 26, 2026

1. What was asked, and what came out

The plan's top-priority workstream asks whether the Space Needle's non-halting can be certified by an **automatic set**: a set I of positive integers, recognised by a finite automaton reading binary digits, with

$$6 \in I, \quad F(I) \subseteq I, \quad I \cap H = \emptyset$$

where F is the one-variable Space Needle map and $H = \{1, 2, 4, 8, \dots\}$ is its halting set. Any such I is a complete, finite, machine-checkable proof that the orbit of 6 never halts. This is the one certificate class that the collection's no-congruence theorems do not already exclude, and nobody has pointed it at an arithmetic Collatz-like map before.

Result (machine-verified). There is no such I with at most **7 states**. The search is exhaustive over all 256,182,290 transition structures of that size (every isomorphism class exactly once) and exact for each: given a structure, whether ANY acceptance labelling works is decided by Horn propagation, not sampled. A stronger, start-free statement also holds: there is no nonempty automatic I with $F(I) \subseteq I$ and $I \cap H = \emptyset$ at all with at most **6 states** — so at these sizes automatic certificates cannot decide *any* start of this map, not only 6.

And the obstruction is dynamical, not a failure to see. The natural worry about a bounded search like this is that small automata simply cannot tell the orbit apart from the powers of 2, in which case the theorem would be about digit complexity rather than about the map. That is measured in Section 6 and it is not what happens: at 3, 4 and 5 states there are 4, 100 and 2,887 structures that *do* separate the orbit from H , and closure destroys every one of them — each with an explicit witness, audited by direct simulation.

2. The identity that makes this computable

Write $x = 2^v(2k + 1)$ with $v = v_2(x)$, so the LSB-first word of x is $0^v 1 \cdot w(k)$. The published map $x \rightarrow x + v + (3/2)(x/2^v - 1)$ is then

$$F(x) = x + 3 \cdot (x \gg (v+1)) + v = (2^{v+1} + 3) \cdot k + (2^v + v)$$

(verified against `needle.step1` for every non-power-of-2 $x < 300,000$). Each valuation branch is therefore an **affine map in k** . That is the whole enabling step: multiply-and-add in base 2 is a finite carry automaton, so for a fixed DFA the set of state pairs (state on x , state on $F(x)$) realised inside branch v is computed *exactly* by a product of three finite components — DFA-state on x , carry, DFA-state on y — with carries bounded by $\max(a_v, b_v)$. Branch $v = 0$ is the odd branch $x \rightarrow (5x - 3)/2$, i.e. $(a, b) = (5, 1)$; then $(7, 3), (11, 6), (19, 11), (35, 20), \dots$

2.1 A local-action lemma (new, and useful beyond WS1)

Because $F(x) = x + 3k + v$, if $3k + v < 2^v$ the three pieces of the sum occupy disjoint bit ranges, so the LSB-first word of $F(x)$ is

$$(v + 3k) \text{ in } v \text{ bits} \cdot 1 \cdot w(k)$$

— the high part k is carried through untouched and only the low block of zeros is rewritten; and then $v_2(F(x)) = v_2(3k + v)$. Proof is the disjointness of the ranges; checked on 29,414 (v, k) pairs as well. It says the map acts *locally on the low digits* whenever the valuation is large relative to the tail, which is also the regime the density workstream (WS2) has to count.

3. Why the search is a decision procedure, not a sampling

component	why it is sound
Search space	Canonical initially-connected DFAs by BFS numbering: each isomorphism class of transition structures appears exactly once. Counts produced by the generator are 1, 12, 216, 5,248, 160,675, 5,931,540, 256,182,290, matching the known initially-connected-DFA counts for a two-letter alphabet — an independent check that the enumeration is complete and non-redundant. Structures with unreachable states are covered by the smaller sizes.
Labelling	For a fixed structure, each realised pair (p, q) is an implication $\text{acc}(p) \Rightarrow \text{acc}(q)$; orbit elements are TRUE units; states of powers of 2 are FALSE units. That is a Horn system, so propagating TRUE from the units gives the least model: the structure admits a certificate iff propagation forces no FALSE unit. Exact, not heuristic.
Convention	I is decided on <i>minimal</i> LSB words — the general meaning of “2-automatic”, with no normalisation. (Demanding a trailing-zero-invariant DFA instead is a special case and would make the theorem weaker.) The product therefore also tracks the y -state just after the last emitted 1, which is where y ’s minimal word ends.
Partial branch budget	Only branches $v \leq V$ are imposed. Those conditions are a subset of full closure, so refutation stays sound: an unsatisfiable subsystem means no certificate exists. (A survivor would be only a candidate, owing the branches $v > V$.) Diagnostics: the branches $v \leq 1$ already suffice at every size tested.
Exactness of the pair sets	branch_pairs and branch_pairs_min are tested against brute-force enumeration of the actual (v, k) instances — equality of sets, not containment, for $k < 20,000$ and several structures — so the product neither invents pairs (which could refute wrongly) nor misses them.

4. Results

states n	structures searched	certificate for the orbit of 6	any nonempty F-invariant avoiding H	runtime
1	1	no certificate	no invariant	0.0 s
2	12	no certificate	no invariant	0.0 s
3	216	no certificate	no invariant	0.0 s
4	5,248	no certificate	no invariant	0.0 s
5	160,675	no certificate	no invariant	0.8 s

states n	structures searched	certificate for the orbit of 6	any nonempty F-invariant avoiding H	runtime
6	5,931,540	no certificate	no invariant	60.3 s
7	256,182,290	no certificate	—	2776.3 s

Both columns are exhaustive at their size. The start-free column is the stronger claim and was run with the cheaper branch budget $v \leq 4$; the orbit column used $v \leq 6$. Every entry is a complete search, so the table reads as a single theorem: **no automatic certificate for this map exists below the stated sizes.**

5. Calibration: the search finds certificates when they exist

An impossibility result from a search is worth exactly as much as the search's ability to succeed. The same code was run on a control machine with the same halting set and the same branch format — $C(x) = 4x$, written as $a_v = 2^{v+3}$, $b_v = 2^{v+2}$ — whose orbit from 6 is 6, 24, 96, 384, ... It finds a certificate at **2 states**: $I = \{x : x \text{ has an even number of 1 bits}\}$, which contains 6, is preserved by $x \rightarrow 4x$ (a shift), and misses every power of 2 (one 1 bit). The certificate is then re-verified by brute force against the actual map over $x < 200,000$, not merely accepted from the search. So the machinery does produce certificates; the Needle simply has none at these sizes.

6. Anatomy of the impossibility: separation versus closure

Two quite different things could kill a structure. Either it cannot **separate** the orbit from H at all — some orbit element and some power of 2 reach the same state, a statement about digit complexity with no dynamics in it — or it separates them and then **closure** drags a power of 2 in. Only the second is about the map. The split, measured exhaustively:

states	structures	separate the orbit from H	survive closure too	killed by closure
3	216	4	0	4
4	5,248	100	0	100
5	160,675	2,887	0	2,887

Every separating structure dies, and dies at the *first* closure step: in all 504 audited cases the witness chain has length one. The mechanism is concrete and always the same shape. Membership depends only on the state a number's word reaches, so if some x sits in the same state as an orbit element, then $x \in I$, hence $F(x) \in I$ — and $F(x)$ turns out to be (or to share a state with) a power of 2. A worked witness, for the 4-state structure $\delta = [[0,1],[1,2],[1,3],[1,2]]$:

orbit element $101 \in I$ reaches state 3 | 7 also reaches state 3, so $7 \in I$ | $F(7) = 16 \in I$ | $16 = 2^4 \in H$ — contradiction.

Each witness is audited independently of the automaton machinery: the integers are re-run through needle.step1 and the words re-run through the DFA. 1,620 witnesses were audited with no failures — 504 on separating structures (all 4 at size 3, all 100 at size 4, 400 of the 2,887 at size 5) and 1,116 on structures drawn without regard to separation (all 216 at size 3, samples of 300 at sizes 4, 5 and 6).

7. What the result means

The real requirement is avoiding the halting basin, not the halting set. A certificate must contain the orbit and be closed, so it must avoid $H^* = \bigcup_j F^{-j}(H)$ — every number whose orbit ever halts, not merely the powers of 2 themselves. That is what the witnesses expose: the length-one chains are exactly first-layer basin elements (halting seeds) sitting in the same automaton state as an orbit element. So the theorem should be read as: **at these sizes no finite-state partition of the integers separates the orbit of 6 from the halting basin.**

It is the automatic generalisation of our no-congruence theorems. A congruence class is the degenerate automatic set — the 1-state-per-residue case. The collection already proved no congruence separates the orbit from H . This replaces “congruence” by “any finite automaton on binary digits”, which is a strictly richer class (it can express digit patterns, not just residues), and the separation still fails. That is the first structural explanation the program has for why regular deciders bounce off cryptids — and it is a different statement from the community’s FAR experience, whose automata run on unary-coded tape languages and collapse to congruence-plus-threshold expressiveness.

The negative half of the two-sided bet has landed; the positive half is still open. The plan predicted exactly this fork. What we have is the bounded impossibility theorem plus the machinery to keep pushing; what we do not have is any evidence that a larger automaton would succeed.

8. Where this goes next

direction	content
Push the size bound	Exhaustive enumeration ends around 8 states (12.7 billion structures). Beyond that, either a SAT encoding (none of pysat/z3/minisat is installed here) or an incremental DFS that assigns transitions lazily and prunes as soon as a constraint’s words are determined. The separating structures are a tiny fraction (2,887 of 160,675 at $n = 5$), so building only separating structures should pay for itself.
The MSB convention	2-automatic is encoding-independent as a class, but the state count is not: an MSB-first automaton can be exponentially smaller. Reversal bounds only give $\text{MSB} \leq 2$ states from the present result. A direct MSB search needs the multiply-add relation read most-significant-first, which is rational but nondeterministic — worth doing, and honestly a gap until it is.
Weighted certificates (the WFAR analogue)	Add an integer weight per transition and accept on an interval. This is how bbchallenge’s WFAR strictly extends FAR, it subsumes both automatic sets and the collection’s affine potentials, and the Horn structure survives if the weights are fixed by search.
Other machines, other bases	The construction only needs branches affine in the tail: machine 3 in base 3 (halting set = powers of 27), the Hydra family in base q , and Fenrir once its case file exists. Cross-machine comparison of the smallest refuted size would be a genuine new hardness measure.

direction	content
The general conjecture	Every nonempty 2-automatic F-invariant meets H. A proof would explain cryptid decider-resistance outright. The witness structure suggests the route: show the halting basin is unavoidable in the automatic topology — i.e. every finite-index digit partition mixes basin elements with orbit elements. Note this cannot follow from counting alone: the basin is thin (about one halting seed per octave), so the argument must be about digit patterns, not density.

9. Verification ledger

claim	status
$F(x) = x + 3(x \gg (v+1)) + v$ and the branch form	PROVED (algebra) and machine-verified against needle.step1 for all non-powers-of-2 below 300,000.
Local-action lemma (Section 2)	PROVED (disjoint bit ranges); also checked on 29,414 (v, k) pairs.
Branch pair sets are exact	MACHINE-VERIFIED: equality against brute-force enumeration of instances, several structures, $k < 20,000$, $v \leq 4$, both conventions.
Enumeration is complete and non-redundant	MACHINE-VERIFIED: generator counts match the known initially-connected-DFA counts through 7 states.
Horn propagation decides a structure exactly	PROVED (least model of a Horn system) — given the pair sets and the units.
No certificate at ≤ 7 states; no nonempty invariant at ≤ 6	MACHINE-VERIFIED by exhaustive search, sound because the imposed branch conditions are a subset of full closure. Not a statement about larger automata.
Witnesses	MACHINE-VERIFIED independently: integers re-run through needle.step1, words re-run through the DFA; 1,620 audited, zero failures.
Anything about certificates of unbounded size	OPEN. Nothing here bounds the general question, and no heuristic in this report should be read as evidence either way.

Code: dfa_invariant.py (machinery and tests), search.py (orbit certificate search), anystart.py (start-free search), witness.py (witness extraction and audit), strength.py (separation versus closure), verify.py (calibration). Every script is runnable standalone and prints its own checks.